

Discussion #10 Solutions

Note: Your TA will probably not cover all the problems. This is totally fine; the discussion worksheets are not designed to be finished within an hour. They are deliberately made slightly longer so they can serve as resources you can use to practice, reinforce, and build upon concepts discussed in lectures, labs, and homework.

Learning Objectives

Bias and Variance

Model selection: Define bias, variance, irreducible error.

Model risk = model bias² + variance + irreducible error.

Overfitting, underfitting: Describe how model complexity affects model bias and variance. Define underfitting and overfitting.

Parameter Inference and Bootstrap

Parameter inference: Correlation and causation. Regression inference with model parameters; confounders.

Bootstrap: Repeated sampling with replacement.

Disclaimer: Think of these learning objectives as a high-level guide to help you organize your studying. This is not a comprehensive list of every topic covered in lecture or discussion.

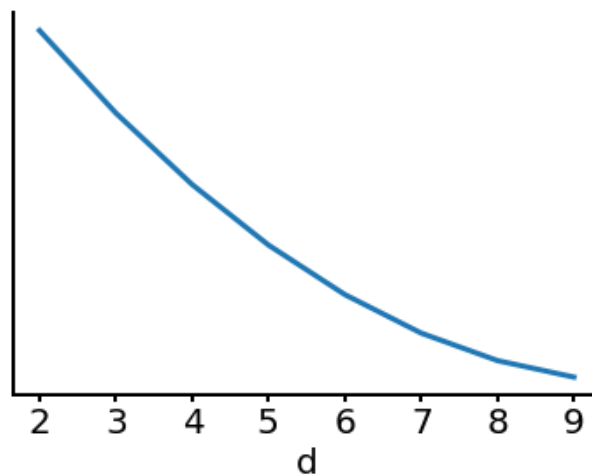
Bias-Variance Tradeoff

1. Your team would like to train a machine learning model to predict the next YouTube video a user will click on based on the videos the user has watched.

We extract up to d attributes (such as length of video, view count, etc.) from each video, and our model will be based on the previous m videos watched by that user. Hence, the number of features for each data point for the model is $m \times d$. Currently, you're not sure how many videos to consider.

Note: given number of users n , the design matrix shape is $n \times (m \times d)$.

- (a) Your colleague, Lillian, generates the following plot, where the value d on the x -axis denotes the number of features used for a particular model. However, she forgot to label the y -axis. Assume that the features are added to the model in decreasing levels of importance: More important features are first, and less important ones are after.



Which of the following could the y -axis represent? Select all that apply.

- ☐ A. Training Error
- ☐ B. Validation Error
- ☐ C. Bias
- ☐ D. Variance

Solution:

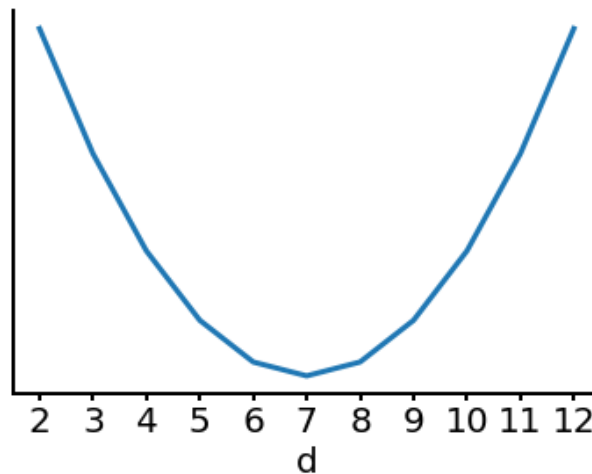
Training Error: Training error can decrease as we add more features.

Validation Error: This can be true depending on the underlying complexity of the data. In part (b), increasing d does not necessarily lower our validation error.

Bias: Decreases with increasing model complexity.

Variance: Increases with increasing model complexity.

- (b) Lillian generates the following plot, where the value d is on the x-axis. However, she forgot to label the y-axis again.



Which of the following could the y-axis represent? Select all that apply.

- ☐ A. Training Error
- ☒ B. Validation Error
- ☐ C. Bias
- ☐ D. Variance

Solution:

We can reason by the process of elimination. Under normal conditions, it can't be the training error since the training error cannot begin to increase when adding model complexity. Bias also decreases with model complexity and variance increases with model complexity. This leaves validation error as the only option, giving $d = 7$ as the optimal number of features to use.

- (c) Explain what happens to the error on the test set as we increase d . Why?

Solution: For smaller d , we are *underfitting* the training data since the model is less complex. The number of features we have does not allow us to fully understand the pattern and so our test error is high. As we increase d , we have more features that allow us to pick up on some of the trends in the data and so the test error decreases a bit.

However, there comes a point after which the features result in overfitting. Our model becomes more complex and does not seem to generalize as well to unseen data resulting in a high test error once again.

2. We randomly sample n data points, (x_i, y_i) , and use them to fit a model $\hat{f}(x)$ according to some procedure (e.g. OLS, Ridge, LASSO). Then, we sample a new data point (independent of our existing points) from the same underlying data distribution. Furthermore, assume that we have a function $f(x)$, which represents the ground truth, and ϵ , which represents random noise. ϵ is produced by some noise generation process such that $\mathbb{E}[\epsilon] = 0$ and $\text{Var}(\epsilon) = \sigma^2$; ϵ is independent of θ .

Whenever we query Y at a given x , we are given $Y = f(x) + \epsilon$, a corrupted version of the real ground truth output. We showed in the lecture that:

$$\underbrace{\mathbb{E}[(Y - \hat{f}(x))^2]} = \underbrace{\sigma^2}_{\text{observation variance}} + \underbrace{(f(x) - \mathbb{E}[\hat{f}(x)])^2}_{(\text{model bias})^2} + \underbrace{\mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]}_{\text{model variance}}$$

- (a) Label each of the terms above using the following word bank. Not all words will be used.
- observation variance
 - model variance
 - (observation bias)²
 - (model bias)²
 - model risk
 - empirical mean square error

Solution:

$$\underbrace{\mathbb{E}[(Y - \hat{f}(x))^2]}_{\text{model risk}} = \underbrace{\sigma^2}_{\text{observation variance}} + \underbrace{(f(x) - \mathbb{E}[\hat{f}(x)])^2}_{(\text{model bias})^2} + \underbrace{\mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]}_{\text{model variance}}$$

- (b) What quantities are random variables in the above equation? In our assumed data-generating process, where is the randomness in each variable coming from (i.e., which part of the assumed underlying model makes each random variable ‘random’)?

Solution: Y : this is the new observation at x . Its randomness comes from the random noise ϵ .

$\hat{f}$: this is the model fitted from the data. Its randomness comes from sampling and the random noise ϵ .

$\hat{\theta}$: these are the optimal theta parameters. Like the model itself, the randomness stems from the sampling and the random noise ϵ .

- (c) Calculate $\mathbb{E} [\epsilon \hat{f}(x')]$, where $\hat{f}(x')$ is a prediction at a fixed data point x' of a model trained on a random sample, and ϵ is the observation error for a new data point at this fixed value of x' .

Hint: Recall that $\mathbb{E}[\epsilon] = 0$.

Solution: Note that $\hat{f}(x)$ is a random variable whose randomness comes from $\hat{\theta}$ being random, which is itself random because of randomness in the training data. The data-generating mechanism for our new data point at x is independent of all the other data used to train our model by assumption, so we have that the sources of randomness for ϵ and $\hat{f}(x)$ are independent. Since ϵ and $\hat{\theta}$ are independent,

$$\mathbb{E} [\epsilon \hat{f}(x)] = \mathbb{E} [\epsilon] \mathbb{E} [\hat{f}(x)]$$

And since $\mathbb{E} [\epsilon] = 0$ by the definition we have given. Therefore,

$$\mathbb{E} [\epsilon \hat{f}(x)] = \mathbb{E} [\epsilon] \mathbb{E} [\hat{f}(x)] = 0$$

Note that ϵ here refers to the noise associated with our new data point x' , which is independently drawn (as stated at the start of the question). This can seem a bit confusing since we've previously used ϵ to refer to the noise associated with each of the data points, but it is used to simplify the notation. It might help to think of the noise associated with each of our points x_i as being ϵ_i (which are i.i.d.), with the noise associated with x' as being ϵ' . It then becomes clearer why ϵ' and $\hat{f}(x)$ are independent.

Regularization and Bias-Variance Trade-off

3. We will use a simple constant model $f_\theta(x) = \theta$ to show the effects of regularization on bias and variance. For the sake of simplicity, we will assume that there is no noise or observational variance, so the ground truth output is equal to the observed outputs: $Y = f(x)$.

- (a) Recall that the optimal solution for the constant model with an MSE loss and a dataset $\mathcal{D}$ with $y_1, y_2, \dots, y_n$ is the mean $\bar{y}$.

We use L2 regularization with a regularization penalty of $\lambda > 0$ to train another constant model. Derive the optimal solution to this new constant model **with L2 regularization** to minimize the objective function below:

$$R(\theta) = \left(\frac{1}{n} \sum_{i=1}^n (y_i - \theta)^2 \right) + \lambda \theta^2$$

Note: As mentioned in the lecture, we do not impose a regularization penalty on the bias term and this problem only serves as a practice.

Solution: We use calculus to derive this result below:

$$\frac{dR}{d\theta} = \left(\frac{1}{n} \sum_i \frac{d}{d\theta} (y_i - \theta)^2 \right) + \lambda \frac{d}{d\theta} \theta^2 = \left(-\frac{2}{n} \sum_i (y_i - \theta) \right) + 2\lambda\theta$$

Set the derivative to 0, and solve for the optimal $\hat{\theta}$.

$$\left(-\frac{2}{n} \sum_i (y_i - \hat{\theta}) \right) + 2\lambda\hat{\theta} = 0 \implies \left(-\frac{1}{n} \sum_i y_i \right) + \hat{\theta} + \lambda\hat{\theta} = 0$$

The optimal $\hat{\theta}$ is:

$$\begin{aligned} \frac{1}{n} \sum_i y_i &= \hat{\theta}(1 + \lambda) \\ \hat{\theta} &= \frac{1}{n(1 + \lambda)} \sum_i y_i = \frac{1}{1 + \lambda} \bar{y} \end{aligned}$$

Note that the magnitude of theta will be smaller in this case since $\frac{1}{1+\lambda} < 1$ for $\lambda > 0$.

- (b) Use the bias-variance decomposition to show that for a constant model **with L2 regularization**, its optimal expected loss on a sample test point (x, Y) in terms of the training data y is equal to the following.

$$\mathbb{E}_{\mathcal{D}}[(Y - \hat{f}(x))^2] = \left(Y - \frac{1}{1 + \lambda} \mathbb{E}_{\mathcal{D}}[\bar{y}] \right)^2 + \frac{1}{(1 + \lambda)^2} \text{Var}_{\mathcal{D}}(\bar{y})$$

What expected loss do we obtain when $\lambda = 0$, and what does that mean in terms of our model?

Note: The subscript next to the expectation and variance lets you know what is random inside the expectation (i.e., what is the expectation taken over?). In this case, we calculate the expectation and variance of $\bar{y}$ across the dataset $\mathcal{D}$.

Solution: Starting from the bias-variance decomposition presented in the lecture:

$$\mathbb{E} \left[(Y - \hat{f}(x))^2 \right] = \sigma^2 + (f(x) - \mathbb{E} [\hat{f}(x)])^2 + \mathbb{E} \left[(\hat{f}(x) - \mathbb{E} [\hat{f}(x)])^2 \right]$$

Recall that we are not dealing with any noise ($\epsilon \sim \mathcal{N}(0, 0)$), so we can eliminate the σ^2 term. Then, $f(x) = Y$, so we can replace $f(x)$ with Y .

$$\mathbb{E} \left[(Y - \hat{f}(x))^2 \right] = (Y - \mathbb{E} [\hat{f}(x)])^2 + \text{Var}(\hat{f}(x))$$

We can then derive $\hat{f}(x)$ using the assumptions of our constant model. We know that $\hat{f}(x) = \frac{1}{1 + \lambda} \bar{y}$. We simplify the following expressions:

$$\mathbb{E} \left[(Y - \hat{f}(x))^2 \right] = (Y - \mathbb{E} \left[\frac{1}{1 + \lambda} \bar{y} \right])^2 + \text{Var}(\frac{1}{1 + \lambda} \bar{y})$$

We can simplify:

$$\mathbb{E} \left[(Y - \hat{f}(x))^2 \right] = (Y - \frac{1}{1 + \lambda} \mathbb{E} [\bar{y}])^2 + \frac{1}{(1 + \lambda)^2} \text{Var}(\bar{y})$$

Setting $\lambda = 0$ is equivalent to using a constant model **without** L2 regularization, and we get the following equation for the expected loss:

$$\mathbb{E} \left[(Y - \hat{f}(x))^2 \right] = (Y - \mathbb{E} [\bar{y}])^2 + \text{Var}(\bar{y})$$

- (c) Remark on how regularization has affected the model bias and variance as λ increases. Consider what would happen to these quantities as $\lambda \rightarrow \infty$.

Solution: The model bias has increased since the gap between a test point Y and a scaled “best estimate” value is likely larger than between the test point and the actual best estimate. As $\lambda \rightarrow \infty$, the model bias squared will tend towards Y^2 .

The model variance has decreased since $\frac{1}{(1+\lambda)^2}$ if $\lambda > 0$ will always be less than 1. Hence, our variance compared to the vanilla bias-variance decomposition from part (b) is reduced by a factor greater than 1. As $\lambda \rightarrow \infty$, the model variance will become 0.

Parameter Inference and Bootstrap

4. A team of researchers is studying the consumption of sugary drinks of adults living in Berkeley. After conducting a background study, the team explores the relationship between consumption of sugary drinks and blood sugar level.
- (a) Suppose we want to use the bootstrap to find a percentile confidence interval for one of the parameters in a model predicting blood sugar level. Which of the following statements are true?
- ☐ A. For a particular 90% confidence interval, there is a 90% chance that the true population parameter is inside the interval.
 - ☒ B. Roughly 90% of several 90% confidence intervals created using the same process should contain the true population parameter.
 - ☐ C. Exactly 90% of several 90% confidence intervals created using the same process should contain the true population parameter.
 - ☐ D. A 90% confidence interval is generally wider than a 95% confidence interval.
 - ☒ E. A 90% confidence interval is generally narrower than a 95% confidence interval.
 - ☐ F. The bootstrapped sampling distribution of an estimator must be normal in order for us to create a bootstrap percentile confidence interval.
 - ☐ G. None of the above

Solution: A 90% confidence interval specifies that we are 90% confident that the true parameter falls between the bounds of our interval. This also means that roughly 90% of similarly constructed confidence intervals across parallel universes will contain the true parameter. However, for a particular 90% confidence interval, the true parameter either falls inside the interval or it does not.

We can still create a percentile confidence interval using our sampling distribution of the estimator across these synthetic parallel universes, regardless of whether or not the distribution is normal.

- (b) To study the relationship, the team fits the following linear regression model over the data:

$$\widehat{\text{blood_sugar}} = \hat{\theta}_0 + \hat{\theta}_1 \text{drink_soda} + \hat{\theta}_2 \text{t2d}$$

- **blood_sugar:** The pre-breakfast blood sugar level measured by the research team in the unit of millimoles per liter. (type: `np.float`)
- **drink_soda:** Whether the surveyed adult consumes at least 1 soda every day. The value is `True` if the surveyed adult consumes at least 1 soda every day, and `False` otherwise. (type: `boolean`)

- **t2d**: An indicator of whether or not the individual has type 2 diabetes, where a value of 1 indicates type 2 diabetes, and 0 indicates no type 2 diabetes. (type: `int`)

1. What is the interpretation of $\hat{\theta}_0$? Explain in one sentence.

Solution: $\hat{\theta}_0$ is the predicted blood sugar for someone who does not drink soda and does not have type 2 diabetes.

2. What is the interpretation of $\hat{\theta}_1$? Explain in one sentence.

Solution: $\hat{\theta}_1$ is the predicted increase in blood sugar for someone who drinks soda relative to someone who does not drink soda, holding type 2 diabetes status constant.

3. What is the interpretation of $\hat{\theta}_2$? Explain in one sentence.

Solution: $\hat{\theta}_2$ is the predicted increase in blood sugar for someone who has type 2 diabetes relative to someone who does not have type 2 diabetes, holding soda drinking status constant.